

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium <i>nonmag.</i>			
$k_i = \frac{\omega\sqrt{\varepsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\varepsilon_{r,2}/\varepsilon_{r,1}}$			
Direction table $u \in \{x, y, z\}$ = axis perp. to boundary			
Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a ₀):	
$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)	(incident wave, Med 1)
$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)	(reflected wave, Med 1)
$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)	(transmitted wave, Med 2)
(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)	
Total in medium 1: $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.	
ê by polarization plug into templates above	

Pol.	ê	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.</i>			
	⊥ pol (θ _i = 0)	pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$	
R refl. pwr	$ \Gamma_{\perp} ^2$	$ \Gamma_{\parallel} ^2$	
T trans. pwr	$ \tau_{\perp} ^2 \frac{\eta_1}{\eta_2}$	$ \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	
R+T total	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$	

Notes: sin θ_t = √μ₁ε₁/μ₂ε₂ sin θ_i; nonmag. ⇒ η₂/η₁ = n₁/n₂.

Intrinsic impedance η_i plug into table above	
$\eta_i = \sqrt{\frac{\mu_r}{\varepsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\varepsilon_{r,i}}} \text{ (}\Omega\text{)}$	$\eta_0 = 120\pi \approx 377 \Omega$
Default μ _r = 1 (air, dielectric). Use μ _r ≠ 1 only if problem says “ferromagnetic”/iron/ferrite/etc.	
Low-loss correction $\eta_c \sigma/(\omega\varepsilon) \ll 1$	
$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon}} \left(1 + j \frac{\sigma}{2\omega\varepsilon}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\varepsilon_r}}$	
For Γ/τ just use η ₀ /√ε _r . The j term is for η _c , θ _η . See LOSSY MEDIA — 5 CASES.	

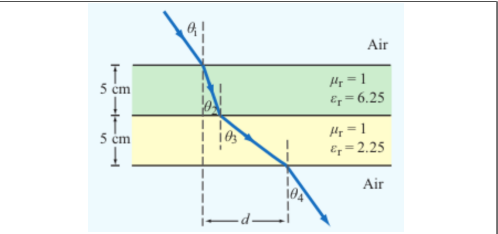
POWER REFLECTION & TRANSMISSION

% reflected <i>always</i> %P _r = 100 Γ ²
% transmitted <i>η-ratio matters!</i> %P _t = 100 τ ² η ₁ /η ₂
Check: %P _r + %P _t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary <i>angles from normal</i>	$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$
Multilayer chain <i>stack of slabs</i> 1 → 2 → 3 → 4	$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\varepsilon_{r,i}}{\varepsilon_{r,i+1}}}$
Air-slabs-air shortcut <i>outer media identical</i>	$\theta_{\text{exit}} = \theta_{\text{incident}}$
<i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m		
	$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$	
Type	σ/ωε	η _c , α, β
Lossless (σ=0)	0	η = η ₀ /√ε _r exact α = 0, β = ω√ε _r /c
Low-loss	< 10 ⁻²	η _c ≈ $\frac{\eta_0}{\sqrt{\varepsilon_r}} \left(1 + j \frac{\sigma \eta_0}{2\omega \varepsilon_r}\right)$ α ≈ $\frac{\sigma \eta_0}{2\sqrt{\varepsilon_r}}$, β ≈ $\frac{\omega \sqrt{\varepsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: (η/√X)e ^{jθ_η} see below
Good cond.	> 10 ²	η _c ≈ (1+j)√πfμ/σ α ≈ β ≈ √πfμσ
Magnetic	any	keep full √μ _r /ε _r η ₀

For Γ/τ: drop j correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params σ/(ωε) ≪ 1, μ _r = 1	
α ≈ $\frac{\sigma \eta_0}{2\sqrt{\varepsilon_r}}$ (Np/m), β ≈ $\frac{\omega \sqrt{\varepsilon_r}}{c}$ (rad/m), η _c ≈ $\frac{\eta_0}{\sqrt{\varepsilon_r}}$ (Ω)	
Exact (quasi-cond) X = √[1 + (ε''/ε') ²]	
α = ω√ $\frac{\mu \varepsilon'}{2}$ √X-1, β = ω√ $\frac{\mu \varepsilon'}{2}$ √X+1	
θ _η = $\frac{1}{2} \arctan \frac{\sigma}{\omega \varepsilon'}$	

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$	TM mode $H_z = 0, E_z \neq 0$
Dimensions a × b with a > b along $\hat{x}, \hat{y}$; wave in $\hat{z}$.	
E _x form (TE) read m, n from arguments	
$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$	
coef. on x = mπ/a; coef. on y = nπ/b.	

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

e^{-jβz} factor omitted. k_c² = (mπ/a)² + (nπ/b)². Ê in V/m, Ĥ in A/m, Z in Ω.

TE mode <i>generator: Ê_z</i>	$\hat{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$
$\hat{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\hat{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\hat{E}_z = 0, \hat{H}_x = -\hat{E}_y/Z_{TE}, \hat{H}_y = \hat{E}_x/Z_{TE}$	
$Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	

TM mode <i>generator: Ê_z</i>	$\hat{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$
$\hat{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\hat{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\hat{H}_z = 0, \hat{H}_x = -\hat{E}_y/Z_{TM}, \hat{H}_y = \hat{E}_x/Z_{TM}$	
$Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	

TEM plane wave <i>for reference</i>	$\hat{E}_x = E_0 e^{-j\beta z}, \hat{H}_y = \hat{E}_x/\eta$
-------------------------------------	---

$$\eta = \sqrt{\mu/\varepsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A, } k = \omega\sqrt{\mu\varepsilon}$$

Properties common to TE/TM	
$f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)}$	
$\beta = k\sqrt{1 - (f_c/f)^2} \text{ (rad/m)}$	
$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}} \text{ (m/s)}$	

CUTOFF FREQUENCY

Common cutoffs (a > b) f _c formula in Table 8-3	
Mode	f _{mn}
u _{p0}	c/√ε _r (hollow: u _{p0} = c)
TE ₁₀	f ₁₀ = u _{p0} /(2a) (dominant)
TE ₀₁	f ₀₁ = u _{p0} /(2b)
TE ₂₀	f ₂₀ = u _{p0} /a = 2f ₁₀
TE ₁₁ , TM ₁₁	f ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$
Propagation requires f > f _{mn} . In Z _{TE} /Z _{TM} and u _g , f _c = f _{mn} of the excited mode.	

ε_r FROM DISPERSION

Given ω, β, mode (m, n) result is unitless	
$\varepsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$ (unitless)	
Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10 ⁸ (m/s), m, n (unitless).	

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}, u_p u_g = u_{p0}^2$
t = L/u _g (s) travel time through guide of length L
Higher modes have lower u _g ⇒ arrive later.

BOUNDARY

η _i , η _t — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = Γ ²
k _i = ω√ε _{r,i} /c (rad/m)
α ₂ (Np/m), β ₂ (rad/m): lossy
z = 0 — boundary plane (m)
η ₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ _i , θ _t — angles from normal
n _i = √ε _{r,i} — index of refr.
t _i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
⊥: E ⊥ this plane
: E this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f _{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z _{TE} /TM — wave imp. (Ω)
E _x — E-field comp. (V/m)
H _y — H-field comp. (A/m)
η = η ₀ /√ε _r

VELOCITIES

u _{p0} = c/√ε _r — bulk phase vel.
u _p = u _{p0} /√[1 - (f _c /f) ²]
u _g = u _{p0} √[1 - (f _c /f) ²]
u _p u _g = u _{p0} ²
t = L/u _g — travel time
c = 3 × 10 ⁸ m/s

POWER & UNITS

%P _r , %P _t — pwr refl./trans. (unitless)
%P _r = 100 Γ ²
%P _t = 100 τ ² η ₁ /η ₂
%P _r + %P _t = 100
σ (S/m), ε (F/m), μ (H/m)
ε ₀ = 8.85 × 10 ⁻¹² F/m
μ ₀ = 4π × 10 ⁻⁷ H/m

DIPOLE TYPE BY LENGTH
 l = antenna rod length (m)

λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
l/λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\ \Omega$
= 1	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199\ \Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$ $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$ Avg power density $l/\lambda < 1/50$; far field	
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta$ (W/m ²)	
<i>Equivalent:</i> $S = S_{\text{max}} \sin^2 \theta$ with $S_{\text{max}} = \eta_0 k^2 I_0^2 l^2 / (32\pi^2 R^2)$ at $\theta = \pi/2$.	
l : rod length (m) R : obs. distance (m) $k = 2\pi/\lambda$ (rad/m)	I_0 : peak current (A) θ : angle from dipole axis (rad) $\eta_0 = 120\pi \approx 377\ \Omega$
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2$ (W)	

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Avg power density any l; far field	
$S(R, \theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$ (W/m ²)	
<i>At $\theta = 0, \pi$: bracket is 0/0. $L'H\hat{o}pital \Rightarrow S = 0$ (no axial radiation). TI-36X alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.</i>	
Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\text{max}}$	

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion multiply by $\pi/180$ or $180/\pi$ $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$						
°	rad	sin θ	cos θ	sin ² θ	cos ² θ	θ^2
0	0	0	1	0	1	
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4	
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2	
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4	
90	$\pi/2$	1	0	1	0	
180	π	0	-1	0	1	

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos \theta$ $F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$	
Normalize $F_a^{\text{norm}} = F_a/N^2$	
HPBW (broadside, uniform) use Nd, not $(N-1)d$ $\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) = $50.8^\circ \cdot \frac{\lambda}{Nd}$	
Electronic steering (scan to θ_0) replace γ with $\gamma' = kd(\cos \theta - \cos \theta_0)$ $\delta = \frac{2\pi d}{\lambda} \cos \theta_0$ ($\frac{\text{rad}}{\text{elt}}$); $\theta_0 = \frac{\pi}{2}$: broadside, 0: end-fire	
Frequency scan feed-line incr. $\Delta \ell = n_0 \lambda_0$ $\cos \theta_0 \approx \frac{n_0 \lambda_0}{d} \frac{\Delta f}{f_0}$	
<i>Grating lobes appear if $d > \lambda$. Avoid.</i>	

DIPOLE

l — antenna length (m) $\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s I_0 — peak current (A) $k = 2\pi/\lambda$ (rad/m) R — obs. dist. (m); θ from axis $\eta_0 = 120\pi \approx 377\ \Omega$ $S(R, \theta)$ — pwr density (W/m ²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless) a — coef. on θ^2 in Gaussian β — HPBW (rad or °) Ω_p — pattern solid angle (sr) $D = 4\pi/\Omega_p$ — directivity ξ radiation eff. (unitless); $G = \xi D$ $D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIS / dB

P_t — power transmitted (W) P_r — power received (W) G_t — transmit gain (linear) G_r — receive gain (linear) $S_r = G_t P_t / (4\pi R^2)$ (W/m ²) $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$ $G = 10^{G_{\text{dB}}/10}$ 3 dB = $\times 2$, 10 dB = $\times 10$ $P_N = k_B T_{\text{sys}} B$ (W)
--

APERTURE

A_p — physical area (m ²) $A_e = \lambda^2 D / (4\pi)$ (m ²) $A_e \approx A_p$ for aperture $D \approx 4\pi A_p / \lambda^2$ (unitless) $\beta \approx 0.88\lambda/\ell$ (rect/sq.) $\beta \approx 1.02\lambda/d$ (circular) $2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

ARRAY

N — number of elements d — spacing (m) $\gamma = kd \cos \theta$ $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$ $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$ $F_a^{\text{norm}} = F_a/N^2$ $\beta \approx 0.88\lambda/(Nd)$ (broadside)

BEAM PARAMETERS — β , Ω_p , D

Normalize always start here $F(\theta) = S(\theta)/S_{\text{max}}$ (unitless) Beamwidth β at threshold X Set $F(\theta_X) = X$, solve for θ_X ; $\beta = 2\theta_X$ (symmetric)		
Threshold	$X = 10^{\text{dB}/10}$	θ_X
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any X	$F(\theta_X) = X$
Pattern solid angle Ω_p recognize pattern, look up $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$ (sr) <i>No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$.</i> <i>$\theta$ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.</i>		
Pattern $F(\theta)$	Ω_p (sr)	
Isotropic ($F=1$)	$4\pi \approx 12.57$	
Hertzian sin ² θ	$8\pi/3 \approx 8.38$	
Half-wave dipole	≈ 7.66	
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_{1/2}^2 / \ln 2$	
2-axis HPBW (aperture)	$\beta_{xz} \cdot \beta_{yz}$	
Other (use integral)	numerical	

TI-36X: [2nd] [d/dx] for $\int_\square^\square$, RAD mode, multiply by 2π .

Directivity D always $D = 4\pi/\Omega_p$; common shortcuts below		
$D = \frac{4\pi}{\Omega_p}$ (unitless)	$D_{\text{dB}} = 10 \log_{10} D$	
WORD TRAP: “direction” = θ_{max} (rad); “directivity” = D (unitless).		
Gain (if asked) ξ = radiation efficiency (unitless), $0 < \xi \leq 1$ $G = \xi D$ (unitless) $G_{\text{dB}} = 10 \log_{10} G$ (dB)		
Lossless / ideal antenna $\Rightarrow G = D$.		
$\xi = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}}$; $R_{\text{rad}} = 2P_{\text{rad}}/I_0^2$ (Ω); $R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}$.		

COMMON DIPOLE PARAMETERS

Lossless ($\xi=1$): $G=D$.				
Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199
Half-wave: $P_{\text{rad}} = 36.6 I_0^2 \Omega$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but P_{rad} , R_{rad} halved.				

ANTENNA AREAS (A_e , A_p)

Effective area antenna's RX cross section; D from BEAM PARAMS or table above $A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)	
Physical cross section wire dipole, diameter d; l see table above $A_p = l d$ (m ² , any dipole)	
Inverse: D from A_e $D = \frac{4\pi A_e}{\lambda^2}$ (unitless)	
<i>Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).</i>	

FRIS TRANSMISSION FORMULA

<i>For each antenna's G: if given by NAME (“half-wave dipole” etc.) $\rightarrow$ COMMON DIPOLE PARAMS ($G=D$, lossless). Else (dB/linear value given) $\rightarrow$ use as given (convert dB $\rightarrow$ linear).</i>	
Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$ $S_r = \frac{G_t P_t}{4\pi R^2}$ (W/m ²); $P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$ (W)	

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$$

Equivalent form using $A_{e, \text{rec. area form}}$
 $P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$, $G_r = \frac{4\pi A_{e,r}}{\lambda^2}$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F=1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain to linear: $G = 10^{G_{\text{dB}}/10}$.

3. Get G_t, G_r by antenna type:

- named dipole $\rightarrow$ COMMON DIPOLE PARAMS ($G=D$ lossless)
 - aperture / dish $\rightarrow$ APERTURE ANTENNAS ($G=D \approx 4\pi A_p/\lambda^2$)
 - arbitrary pattern $F(\theta) \rightarrow$ BEAM PARAMS ($G = \xi \cdot 4\pi/\Omega_p$)
 - given in dB $\rightarrow$ dB $\leftrightarrow$ LINEAR table
4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity (G , D , P_r/P_t , SNR , ...): $X_{\text{dB}} = 10 \log_{10} X_{\text{lin}}$ $X_{\text{lin}} = 10^{X_{\text{dB}}/10}$ Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.	
For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).	

NOISE & SNR

Noise power thermal noise into matched receiver $P_N = k_B T_{\text{sys}} B$ (W) $k_B = 1.38 \times 10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).	
Signal-to-noise ratio $\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$	

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m ²); A_e effective area (m ² , $= \lambda^2 D / (4\pi)$); β_{xz}, β_{yz} HPBW's in two planes (rad); ℓ , d aperture side / diameter (m).	
Far-field req.: $R \geq 2d^2/\lambda$ (d = largest aperture dim.).	
Pattern (rect., uniform illum.) peak at $\theta=0$ (broadside) $F(\theta) = \text{sinc}^2 \left(\frac{\pi \ell_x \sin \theta}{\lambda} \right)$; $S_0 = \frac{E_0^2 A_p^2}{2\eta_0 \lambda^2 R^2}$	
Directivity (any shape, uniform illum.) $D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) $\Rightarrow A_e \approx A_p$	
Directivity from beamwidths any aperture $D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}}$ (use rad); circular: $D \approx 4\pi/\beta^2$	

HPBW and D by aperture shape uniform illum.; β in rad			
Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x \beta_y)$ $= \frac{4\pi \ell_x \ell_y}{\lambda^2}$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx \frac{4\pi}{\beta^2} = \frac{4\pi \ell^2}{\lambda^2}$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx \frac{4\pi}{\beta^2} = \frac{\pi d^2}{\lambda^2}$

Scaling (any ratio) $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$

$$D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}} \right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$$

If a parameter is unchanged, its ratio = 1. Examples: double area $\rightarrow D \times 2$, $\beta \times 1/\sqrt{2}$; double freq $\rightarrow D \times 4$, $\beta \times 1/2$.